SUPPLEMENTAL MASTER PROMPT
INDEPENDENT FORENSIC AUDIT, ADVERSARIAL RECONSTRUCTION, AND EVIDENCE-BOUNDED ELEVATION OF THE EXISTING TRADING-SYSTEM BLUEPRINT

Engagement type: Design-only reassessment and reconstruction in an offline, closed-world environment
Execution profile: Arena.ai Direct mode; no internet, browser, network, external connectors, or remote tools
Governing source: The complete, owner-supplied `master prompt sent in 2 parts` (the original master prompt)
Candidate under review: The complete architecture/blueprint previously produced by you in this conversation
Exchange-specific evidence sources: Original `master prompt sent in 2 parts` ratified findings within their stated scope, plus the `TOOBIT FACT PACK` appended verbatim at the end of this prompt; no outside sources
Output language: Professional English Markdown for the design package; Persian for owner-facing progress notes and the short final note, consistent with the original master prompt.

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0. Purpose, authority, and scope of this supplemental prompt

You previously produced an architecture/blueprint in response to the original `master prompt sent in 2 parts`. Your task now is to conduct the most rigorous, broad, independent, adversarial, and evidence-bounded reassessment that your actual available capabilities permit, then correct and reconstruct that blueprint into the strongest defensible design that can be justified by the original master prompt, the later owner decisions explicitly recorded below, and the appended Toobit Fact Pack.

This document is a **supplemental refinement directive**, not a replacement for `master prompt sent in 2 parts`. The original master prompt remains fully operative. Every clause in it remains binding unless this supplemental directive expressly and narrowly updates that clause. Do not silently weaken, reinterpret, omit, or replace any original requirement.

The previous blueprint is a **candidate artifact to be challenged**, not an authority, not evidence that its own claims are true, and not a template that must be preserved. You are not being asked merely to polish wording, add missing headings, or defend your earlier decisions. You must be willing to redesign, replace, split, merge, remove, or reverse any part of the previous architecture when the requirements and evidence justify doing so. Equally, do not discard a sound design decision merely to make the revision look different.

The objective is not to claim literal perfection, universal optimality, guaranteed profitability, or zero uncertainty. The objective is to produce the highest-quality **defensible and internally coherent design** obtainable within the actual evidence, tools, time/context limits, and design-only boundary. Demonstrate quality through traceable requirements, sound derivations, correct calculations, explicit alternatives, falsifiable acceptance criteria, coherent state and failure models, and complete disclosure of what remains unverified.

The final design package must be suitable for a formal architecture review and a later engineering handoff **as a design specification only**. It must not claim that implementation, backtesting, paper operation, exchange connectivity, production deployment, or live trading has been performed or approved.

0.1 Required source inputs

Use these inputs, and distinguish their roles precisely:

1. **Original `master prompt sent in 2 parts`:** the governing requirements, constraints, evidence record, design principles, precedence rules, and delivery obligations. Read the complete file if it is available. Do not rely on its filename, a partial extract, or memory of it as proof that its complete contents were inspected.
2. **Your complete previous blueprint:** the candidate to audit. Inspect the entire available artifact, including appendices, tables, formulas, diagrams, registers, and any continuation messages. Do not assume that a heading implies the relevant behavior has actually been specified.
3. **This supplemental directive:** a limited set of later owner decisions and additional instructions for this refinement engagement. Apply its expressly stated overrides only within their declared scope.
4. **Original `master prompt sent in 2 parts` ratified findings and the appended `TOOBIT FACT PACK`:** these are the only permitted Toobit-specific evidence sources in this engagement. Preserve the original prompt's explicitly ratified findings within their stated scope and apply its precedence rules. Use the Fact Pack as the supplied extract for additional endpoint and exchange details, preserving every item's stated status and limitations. The Fact Pack does not silently revoke an original ratified finding; neither source authorizes adding outside Toobit facts.

If any required input is genuinely inaccessible, incomplete, or truncated, state exactly what is unavailable, mark the dependent work accordingly, continue every independent task that can responsibly be completed, and do not invent the missing content. Do not claim to have inspected a file or section that you did not actually inspect.

0.2 Narrowly defined precedence and amendments

Apply the precedence framework in `master prompt sent in 2 parts` §2. This supplemental directive makes the following later owner declarations explicit:

- The six `D-RISK` decisions in §1 below supersede the intentionally pending risk-boundary values in original `master prompt sent in 2 parts` §3.3. They also refine the capital-planning scale in §3.1 item 11. They do **not** supersede unrelated provisions of the original prompt.
- The Telegram-controlled `Trade → Paper / Live` selector is confirmed. There must be no arbitrary elapsed-time requirement or profitability/performance hurdle for selecting Live rather than Paper. The six safety interlocks listed below and in the original prompt remain mandatory, non-bypassable correctness gates.
- Cross-venue data is restricted to research and historical backtesting in this engagement. It must not be accessible to Paper or Live data, decision, or execution paths, even where a venue might otherwise satisfy the broader conditions in original §3.1 item 5. Preserve source-tag lineage and enforce the separation architecturally, not only by documentation.
- External legal, sanctions, jurisdictional, and broad regulatory determinations are not delegated to you. Do not render legal conclusions. The design's compliance/control scope is limited to applicable Toobit terms and API constraints as represented in the permitted evidence, general security controls, and internal auditability. Any unresolved legal or licensing question remains an owner/legal matter.
- The current task is a design reassessment. Actual implementation, exchange-account access, authenticated calls, backtest execution, Paper execution, Live execution, production deployment, and operational-readiness certification are outside this engagement. Design-level calculations, static consistency checks, schema checks, deterministic reasoning checks, and test specifications remain in scope when the required tools and inputs are actually available.

Where this supplemental directive does not explicitly amend the original master prompt, the original wording controls. If two binding requirements still conflict, expose the conflict and its consequence; do not silently pick a convenient interpretation.

0.3 Arena.ai Direct-mode offline execution profile — controlling instruction

This engagement is intended to run in **Arena.ai Direct mode**, not Agent mode. Treat the following as a hard execution constraint and as a narrow, explicit amendment to any original clause that requires, anticipates, or suggests external research, browsing, or remote source inspection.

1. **Closed-world evidence only.** Work exclusively from the complete materials already present in the conversation or explicitly attached/provided to you: the original `master prompt sent in 2 parts`, the previous blueprint, this supplemental directive, the owner decisions in §1, and the embedded Fact Pack. You may also use your general mathematical, statistical, engineering, and domain knowledge, but identify it as general knowledge or explicit reasoning rather than as newly verified external evidence.
2. **No internet or remote access.** Do not browse, search the web, open or visit URLs, resolve domains, make HTTP/HTTPS/WebSocket/API calls, ping remote hosts, download files, query public documentation, retrieve live market data, access Toobit, access Telegram, or use external plugins, connectors, agents, or remote tools. Do not attempt these actions indirectly through another mechanism.
3. **URLs are provenance metadata only.** Any URL, domain, endpoint documentation address, or external reference appearing in `master prompt sent in 2 parts`, the previous blueprint, or the Fact Pack is descriptive source metadata or a future evidence locator only. It is **not** an instruction or permission to open, visit, query, or fetch it. In particular, do not open any address mentioned in the Fact Pack.
4. **Do not request a mode change.** Do not ask the owner to switch to Agent mode, enable browsing, connect an account, supply credentials, or activate external tools. Do not pause merely because online verification is unavailable.
5. **No external verification claims.** Do not create citations, source quotations, page numbers, endpoint behavior, current exchange facts, or test results that are not in the supplied materials. A Fact Pack item marked `SOURCE_VERIFIED` retains that supplied status as a property of the Fact Pack; it does not mean that you independently re-verified it in this Direct-mode session. State this distinction whenever material.
6. **Offline calculations remain in scope.** Perform calculations, arithmetic checks, schema reasoning, cross-reference checks, state-transition analysis, threat modeling, and other deterministic or analytical work from supplied inputs. If a genuinely local calculation or file-inspection facility is exposed and usable without network access, it may be used; do not assume such a facility exists and do not download packages or dependencies. If no such facility exists, show reproducible manual calculations and state that no executable test was run.
7. **Continue despite missing remote evidence.** If a design question would ordinarily require an external source or live test, do not guess and do not stop the entire task. Isolate the dependent claim or feature, assign the appropriate evidence/test status under §3.3.1, identify the exact evidence artifact or measurement needed later, specify a safe interim design, and continue all independent work.
8. **Override external-research workflow only.** The no-internet restriction suspends the original prompt's requirements to conduct new external-source research in this engagement (including external source searches anticipated by §2.1.3, §4.3, §7, and G1). It does **not** suspend requirements to inspect the supplied files, reconstruct requirements, conduct quantitative/statistical research design using supplied evidence, challenge assumptions, compare architecture options, or perform offline analytical verification. Convert the original pre-research checkpoint into an **offline pre-review checkpoint** based only on supplied materials; do not falsely call it an externally researched or externally verified checkpoint.
9. **Resolution paths must be offline-safe.** When external evidence is missing, describe the future evidence required in concrete terms (for example, a dated document export, an owner-supplied response sample with secrets removed, or a controlled measurement record). You may identify the relevant documentation page or endpoint as a locator, but you must explicitly say that it was not opened or verified in this engagement. Do not word the resolution path as an action you are about to perform.
10. **No fabricated tool inventory.** Your capability inventory must report only what is observable in this

conversation. If web access, code execution, filesystem access, agents, or independent reviewers are not exposed, state `NOT AVAILABLE IN THIS SESSION` or `NOT OBSERVED`; do not infer hidden capabilities from the model brand or Arena mode.

This offline profile has priority over any generic instruction in the original prompt or this supplement that might otherwise be interpreted as requiring web access or remote retrieval. It does not change the underlying project requirements or permit unsupported Toobit claims.

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1. Later owner decisions — binding risk and operating constraints

Treat the following as **OWNER_HARD** decisions. They are not suggestions, examples, calibration targets, optimizer variables, or matters to be reopened. Do not ask the owner to reconfirm them. Do not weaken them to make a design easier. Do not silently replace them with values from the previous blueprint or with industry conventions.

1.1 D-RISK-1 — Maximum peak-to-trough drawdown

The maximum permitted decline from peak equity to subsequent trough equity is **15% of initial equity**. This is a hard limit. **20% is advisory context only** and must never be used as the operative limit, a tolerated breach, an optimizer target, or a substitute for 15%.

Preserve the owner's denominator: initial equity. State the exact equity measure, high-water-mark construction, treatment of unrealized PnL, fees, funding, deposits/withdrawals, and measurement/update timing. Do not silently substitute the conventional percentage drawdown denominator of the prior peak for the owner's stated initial-equity denominator. Where operational details are not established, choose a conservative design interpretation, label it `DESIGN_JUDGMENT` or `UNRESOLVED` as appropriate, state its consequence, and continue.

1.2 D-RISK-2 — Per-trade risk ceiling

The maximum risk budget for any individual trade is **5% of current equity**. This is a ceiling, not a target and not a recommended default position size. The strategy or sizing logic may use less; it must never authorize more. No optimizer, calibration process, adaptation process, portfolio allocator, or operator command may relax this ceiling.

Define precisely how trade risk is estimated, including stop distance, contract multiplier, quantity rounding, fees, spread, slippage, funding over the intended holding horizon, partial fills, gap/market-discontinuity risk, and protective-order failure. Distinguish a modeled or stressed risk budget from a guaranteed maximum realized loss. A stop-price calculation alone must never be represented as a guaranteed loss cap. If essential risk inputs are missing, stale, contradictory, or unverifiable, the risk-increasing action must fail closed.

1.3 D-RISK-3 — Capital-planning basis

Use **200 USDT** as the hard planning basis for design calculations. Perform sensitivity analysis across **100–500 USDT**. This is a design-planning scale, not a statement of the owner's current account balance, not authorization to access an account, and not proof that any minimum order, strategy, fee structure, or deployment is feasible.

Use USDT for all monetary quantities. Show formulas, units, assumptions, and boundary cases. Where the design's data or contract terms do not establish feasibility at a given equity level, mark it `UNRESOLVED`, `BLOCKED`, or `NOT TESTED` as appropriate; do not invent exchange filters or claim feasibility.

1.4 D-RISK-4 — Aggregate simultaneous risk ceiling

The sum of risk across all concurrently open positions and any risk-increasing pending orders must be **no greater than 15% of equity**. This is a hard aggregate ceiling, not a target. Define the equity denominator and aggregation method. Do not net away offsetting positions or correlated exposures unless a demonstrably valid risk model justifies that treatment and the owner limit remains satisfied under the conservative measure.

Pending orders, partially filled orders, multiple positions, correlated instruments, and simultaneous signals must be accounted for so that concurrency or race conditions cannot exceed the ceiling. If aggregate risk cannot be computed reliably, do not add risk.

1.5 D-RISK-5 — Daily realized-loss veto

The maximum daily realized loss is **10% of equity**. Upon reaching or breaching this threshold, the system must veto further risk-increasing activity until the end of that UTC day.

Define the UTC day boundary, daily reference-equity denominator, realized-PnL accounting, and treatment of fees, funding, commissions, rebates, corrections, and late-arriving exchange events. The denominator was not specified more narrowly by the owner; do not ask an avoidable question. Select and justify a conservative operational definition, label the definition as a `DESIGN_JUDGMENT` if it is not directly established, and

state the consequence of the choice.

The daily veto must not prevent actions whose purpose is to reduce risk or protect existing positions, where those actions are supported by confirmed exchange semantics and safe preconditions. Do not invent an exchange-side `reduceOnly` or emergency-order capability absent evidence. If the safe risk-reduction route cannot be established, disclose the limitation and fail safely. The daily veto resets only at the next UTC day boundary; define how other independent safety latches interact with that reset.

1.6 D-RISK-6 — Effective leverage ceiling

Maximum effective leverage is **5×**. This is a hard ceiling, not a desired leverage setting or utilization target. Define effective leverage mathematically, including gross exposure, equity denominator, pending orders, and hedge-mode positions. Do not confuse effective portfolio leverage with the leverage setting accepted by the exchange. The actual per-symbol exchange limit may be lower; the permitted value is never higher than the lesser of the owner's 5× ceiling and the symbol's verified exchange constraint.

Where symbol-specific leverage tiers or margin semantics are not established by the Fact Pack, preserve that uncertainty. Do not infer a universal exchange maximum from an example or from another symbol.

1.7 Interaction among the six hard risk decisions

Treat all six decisions as simultaneously binding. Analyze their interactions, including at least:

- per-trade risk versus aggregate concurrent risk;
- aggregate risk versus peak-to-trough drawdown;
- daily realized loss versus total equity drawdown, including unrealized PnL;
- leverage versus notional exposure, margin, liquidation, and stop-risk;
- equity decline and the resulting change in current-equity-denominated limits;
- pending orders and concurrent order-submission races;
- fees, funding, slippage, partial fills, gaps, exchange/API failure, and inability to cancel or close;
- reset, latch, re-arm, and operator acknowledgement behavior.

Do not assume the numerical limits alone guarantee that realized losses cannot exceed them. Define the risk model's limits, model error, execution gaps, and fail-safe behavior explicitly. No subsystem that forecasts, calibrates, optimizes, allocates capital, or handles Telegram input may override the risk authority.

For every D-RISK item, preserve the exact owner value in one canonical risk-policy catalog and reference that catalog everywhere else. Avoid duplicate or subtly different definitions. These limits must not be optimized, learned, relaxed, or self-adjusted. Original D17 remains in force for every other risk parameter and every calculation that depends on equity, stop distance, costs, contract terms, or other uncertain inputs: show those relationships parametrically, with units and sensitivity. Sensitivity analysis may vary the planning equity across 100–500 USDT and may vary genuinely uncertain model inputs, but it must never vary the six D-RISK hard limits as candidate values or imply that another threshold is authorized.

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2. Non-negotiable project invariants carried forward from master prompt sent in 2 parts

The following are restatements for audit visibility, not replacements or a complete transcription of the original. The complete original prompt remains controlling:

1. **Live venue/instrument:** Toobit USDT-margined futures/perpetual futures. Do not reopen the venue decision without a material contradiction or explicit owner authorization.
2. **Operator interface:** Telegram bot is the only operator-facing interface. Do not introduce a second bot, dashboard, web console, desktop client, mobile app, or routine operator interface. Internal services and monitoring components are permitted only where justified and must not become a parallel operator interface.
3. **Time:** UTC is authoritative for computation, storage, logs, alerts, reports, scheduling, exchange communication, and audit events. Define timestamp semantics and use monotonic time for elapsed deadlines.
4. **No account access:** Do not request secrets, connect to an account, perform authenticated calls, change API permissions, access funds, or place/cancel/amend orders. This is a design-only engagement.
5. **Cross-venue firewall:** Under this supplemental directive, any cross-venue dataset is research/backtest-only. It must be tagged at ingestion and prevented from entering Paper or Live paths by enforceable architectural controls.
6. **Trading intent:** Intraday. Derive an operational definition and holding-duration policy from the proposed strategy, costs, funding, and risk; do not invent a duration merely to complete a table.
7. **Universe:** Derive the usable universe through the original D14 process. The ten C19 measurement symbols are not a universe recommendation or candidate list.
8. **Reference device and deployment:** Preserve the POCO X6 reference device and the complete D15 envelope-first process. Derive compute/storage/latency needs first; enumerate every topology the requirements actually permit before applying the owner's compromise order. Do not let mobile convenience predetermine the architecture.
9. **Telegram reliability:** No stable inbound connection from the handset may be assumed. Telegram loss must not disable safety or make uncertain exchange state appear confirmed. Specify open-position and pending-order behavior when the operator is unreachable.
10. **Phases:** Derive phase count and boundaries from risks and verification dependencies per original D16.

The Paper/Live selector has no arbitrary wait or profitability hurdle, but the six interlocks remain mandatory:
    - data validity;
    - exchange/internal-state reconciliation;
    - all hard risk limits;
    - credential and permission validity;
    - configuration integrity;
    - clock validity.
11. **Delegated technical choices:** Architecture, languages, frameworks, runtimes, data stores, algorithms, model classes, and optimizer choices remain delegated to you under the original prompt. Do not ask the owner to select technical solutions.
12. **Original statistical constraints:** Preserve the original methodology, C20–C24, trial-budget rules, D19, and the ban on exhaustive permutation grid search at timeframes ≤12h. The hard D-RISK decisions above are not optimizer parameters.
13. **Original evidence and archive constraints:** Preserve all original C-series, A-series, D-series, E-series, P-series, G-series, T-C-series, and V-series scope/status distinctions. Do not upgrade an owner summary or a partial archive record into direct source verification.
14. **No implementation/live approval:** The final package is a design artifact. It is not a claim of profitability, successful backtesting, implementation correctness, Paper readiness, production readiness, or authorization for Live trading.

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3. Source perimeter and evidence integrity for this engagement

3.1 Closed-world source perimeter — no internet, no domains, no remote calls

The current execution environment is Arena.ai Direct mode. External internet access is intentionally outside the permitted source perimeter. The prohibition is absolute for this task: do not browse, search, open or visit URLs, resolve domains, call remote websites, invoke REST/WebSocket/API endpoints, query undocumented endpoints, download files, retrieve live data, or use external plugins, connectors, agents, or remote tools. Do not ask to switch to Agent mode or otherwise change this execution profile.

The only permitted project-specific evidence is: (a) the complete supplied `master prompt sent in 2 parts`, including its explicitly ratified findings and evidence/status distinctions; (b) the previous blueprint actually present in the conversation or supplied as a file; (c) the owner decisions in §1; and (d) the embedded `TOOBIT FACT PACK`. No outside Toobit-specific facts may be added. Apply the original prompt's precedence rules and §3.3 below when the original ratified findings and the Fact Pack differ.

The Fact Pack is a dated, static evidence snapshot, not a live contract. Preserve its statuses exactly. `SOURCE_VERIFIED` means the Fact Pack itself labels the item as source-verified; it does not mean you re-opened or independently re-verified the cited source in this session. Do not upgrade any status. URLs and domains within the supplied materials are provenance references or future evidence locators only; never open them. Any Fact Pack wording such as “the implementation must fetch at startup” describes a requirement for the future system being designed; it is not an instruction for you to perform a fetch, call an endpoint, or access the exchange in this engagement.

General mathematical, statistical, quantitative-finance, engineering, software, security, and operational knowledge may be used where relevant. Distinguish such general knowledge from project-specific evidence. A remembered general principle is not proof that Toobit implements a particular endpoint, field, status, order type, margin mode, or API behavior.

3.2 Evidence-pointer requirements

Every material Toobit-dependent claim, interface field, limit, status, protocol, endpoint, behavior, or design decision must point to the exact Fact Pack section/table/item and preserve the status stated there. Where relevant, also cite the original `master prompt sent in 2 parts` C-series finding. For each such claim, state its design consequence.

Do not:

- upgrade `PROVISIONAL`, `UNRESOLVED`, or `DOC_CONFLICT` to `SOURCE_VERIFIED`;
- convert an example response or partial symbol list into a universal contract;
- invent missing endpoint paths, parameters, response fields, negative error codes, status enums, order flags, contract filters, timeframes, leverage tiers, or margin behavior;
- describe a fact as current beyond the Fact Pack's stated extraction date and scope;
- state that an endpoint or capability was tested when it was only listed in the Fact Pack;
- cite a generic Fact Pack heading when a more precise row, field, or C-series finding is available.

For a missing Toobit fact, use `UNRESOLVED`, state why it matters, the exact future evidence artifact or measurement needed to resolve it, the affected design decision, the safe interim behavior, and the consequence of leaving it unresolved. The resolution path must be phrased as a later owner-supplied evidence/measurement requirement, not as a web search or retrieval action in this session. If the evidence would ordinarily come from a named documentation page, identify that page only as a locator and explicitly mark it `NOT INSPECTED IN THIS DIRECT-MODE ENGAGEMENT`. Do not fill the gap with a plausible guess.

3.3 Precedence for original findings and Fact Pack conflicts

Apply original `master prompt sent in 2 parts` §2 precedence. In particular:

- Original owner-ratified findings remain ratified within their recorded scope and are not silently displaced by a newer partial extract.
- Where the Fact Pack itself records a `DOC_CONFLICT`, retain both branches and apply the explicitly stated conservative branch.
- For C1/C2, preserve the original ratified 14-timeframe capability ceiling and the 3,500-candle cap as described in the Fact Pack. The shorter WebSocket interval list is not proof that the REST API supports only those intervals. Use the conservative 1,000-kline response limit where the endpoint conflict remains unresolved. Treat 3,500 as a cap, not a guarantee of availability.
- For C4, use conservative fixed-weight rate accounting as the safe baseline when rate-limit headers are absent. Header-derived deltas may be an optional enhancement only if their presence and semantics are actually observed; do not depend on them.
- For C13, retain the conflicting `markPrice` / `klines.change` semantics and do not build a critical decision on an unverified interpretation. Choose a safe branch or exclude the affected feature pending confirmation.
- For C6, use the Fact Pack's conservative 0.06% taker fee default and preserve 0.05% as unresolved sensitivity, not as settled truth.
- For C7, do not treat the observed −30 bp clamp as a universal funding bound. Funding is a tail-risk/cost input, not a guaranteed capped cost.
- For C3 and per-symbol leverage, do not infer a universal exchange maximum. The owner's 5× effective-leverage ceiling remains independently binding.

If you identify another conflict not explicitly resolved above, record both sources and statuses, apply original §2 precedence, and choose the conservative branch where safe. Do not silently amend the Fact Pack.

3.3.1 Offline verification and status semantics

Keep source status and execution/test status separate; they answer different questions. Use the following rules consistently:

- `SOURCE_VERIFIED` may be retained only when it is explicitly assigned by a supplied evidence record/Fact Pack item. Describe it as **supplied-source status**, not as independent verification performed now.
- `RATIFIED_FINDING` remains an owner-ratified finding from the original prompt within its stated scope and limitations; it is not silently converted into independently inspected raw evidence.
- `PROVISIONAL` remains provisional even when it is useful for a design sensitivity case.
- `UNRESOLVED` describes the underlying fact, semantics, or decision that is not established by the supplied evidence.
- `BLOCKED` describes a verification or action that cannot be completed in this engagement because its required remote source, access, resource, or authorization is outside the permitted environment.
- `NOT TESTED` describes a test, measurement, integration, or operational exercise that has not actually been executed.
- `DESIGN_JUDGMENT` is a reasoned design choice made under incomplete evidence; show the positive reasoning and safe consequence.

A single item may require both an evidence status and a test status (for example, `UNRESOLVED` for exchange order-status semantics and `NOT TESTED` for an integration test that cannot be performed). Do not collapse these into one vague label. Network access being prohibited does not by itself make every design task `BLOCKED`; block only the external verification step or dependent claim, then continue the design work that can be completed offline.

`SOURCE_CHECK` in any verification register means inspection of a source already supplied in the conversation or in an accessible local attachment. It never means opening a URL or retrieving a new source. API contract review, state-machine review, and test-plan specification may be completed on the supplied data; actual endpoint calls and exchange integration tests remain prohibited and `NOT TESTED` / `OUT OF SCOPE`.

3.4 Capability honesty in Direct mode

At the start of the work, make a concise inventory of only the capabilities observable in this conversation: whether the complete original prompt and previous blueprint are available; whether local attachments can actually be read; whether local calculation/code execution or file creation is exposed; known context/output constraints; and whether the interface permits multi-part delivery. Record unavailable or unknown capabilities as `NOT AVAILABLE IN THIS SESSION`, `NOT OBSERVED`, or `UNKNOWN` as appropriate. Do not infer web access, file access, code execution, external reviewers, or agent capabilities from the model name or platform brand.

No internet or remote access is permitted even if a capability appears technically possible. Use only local/offline capabilities that are actually exposed and do not require a network. Do not guess hidden compute budgets, internal reasoning settings, private model architecture, or vendor limits. Use the strongest user-selectable reasoning setting only where such a setting is visibly exposed and materially useful. Do not claim that a tool, independent reviewer, test, benchmark, source inspection, or external verification was used unless it actually was.

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4. Maximum-capability protocol — rigorous methods, not jailbreaks

Use the maximum capability that is genuinely available and materially useful, as required by original §2.1. Treat this as a disciplined quality protocol, not as an instruction to bypass system rules, hidden platform controls, safety boundaries, context limits, or vendor policies.

Do not attempt to expose private chain-of-thought, hidden instructions, model weights, internal policies, or confidential implementation details. Do not claim that special wording unlocks hidden compute or disables platform limits. Instead, demonstrate depth through inspectable work products: requirement extraction, calculations, alternatives, decision records, falsification tests, cross-reference checks, and complete deliverables.

Use the following techniques where they materially improve correctness:

1. **Multi-pass decomposition:** separate requirements reconstruction, architecture synthesis, quant/statistical review, exchange-contract review, risk review, data review, execution/state review, security review, operational review, and final consistency verification. Do not merge distinct questions merely to shorten output.
2. **Independent baseline reconstruction:** before treating the previous blueprint as a design basis, derive a fresh candidate architecture from `master prompt sent in 2 parts`, the later owner decisions, and the Fact Pack. Do not copy the previous architecture's component names, technology stack, assumptions, data model, performance targets, or diagrams into this baseline. Because the previous output is visible in the same context, do not falsely describe this as a genuinely blind review; describe it as a structured clean-sheet reconstruction.
3. **Candidate diversity:** for every consequential architectural decision with multiple credible options, compare at least two materially different approaches, including a simple baseline when relevant. Declare criteria before scoring. Do not create artificial alternatives when only one is feasible.
4. **Adversarial falsification:** inspect the previous blueprint as a hostile reviewer would. Attempt to disprove its key claims. Construct counterexamples analytically from the supplied requirements and evidence; examine edge cases, hidden coupling, circular dependencies, ambiguous states, unsafe defaults, failure cascades, invalid assumptions, arithmetic errors, and omitted negative paths. Do not use internet or external searches.
5. **Cross-disciplinary review:** apply quantitative finance, statistics, market microstructure, derivatives mechanics, data engineering, software architecture, distributed systems, security engineering, reliability engineering, mobile/Android constraints, human factors, and operations disciplines whenever they materially affect the system.
6. **Formal-method techniques where justified:** use invariants, state-transition analysis, preconditions/postconditions, safety properties, liveness properties, dependency graphs, threat models, failure-mode analysis, fault trees, sequence diagrams, property-based test specifications, and contract/schema checks where they answer a real question. Do not add formalism merely for appearance.
7. **Quantitative reproduction:** independently recompute every material equation, ratio, capacity estimate, rate-limit budget, risk calculation, storage estimate, concurrency bound, and completeness count. Show units, intermediate values, assumptions, rounding, and the exact arithmetic. Use executable calculations when a suitable tool is available.
8. **Change-impact propagation:** a correction to one component must be propagated to every affected interface, state machine, table, diagram, risk control, test, requirement, runbook, and acceptance condition. A finding is not resolved merely because a paragraph says it was fixed.
9. **Regression review:** after each material correction, check for regressions against all related hard constraints and dependent requirements. Recompute any affected counts and formulas.
10. **Effort by expected value:** spend the deepest analysis on high-impact, difficult-to-reverse, safety-critical, or weakly evidenced decisions. Do not waste effort repeating settled evidence or adding decorative sophistication.

In this Arena.ai Direct-mode engagement, assume no separate reviewer agent, independent model call, remote specialist, or external connector is available unless it is explicitly exposed within the current conversation. Do not attempt to invoke one or request a mode change. Perform distinct discipline-specific review passes yourself; label them as internal analytical passes, not independent external reviews. A genuinely local/offline tool may be used only if visibly available and materially useful.

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5. Mandatory reassessment workflow

The following is a minimum quality workflow, not a rigid linear execution order. Follow original D11 and derive the actual dependency graph first. Reorder, split, merge, or run workstreams in parallel when dependencies justify it. Record the reason for every consequential deviation.

Pass A — Input and boundary integrity

- Confirm access to the complete original prompt and complete previous blueprint.
- Inventory actual capabilities and prohibitions.
- Freeze the closed-world source perimeter: original prompt and its ratified findings, this owner addendum, prior blueprint as candidate, and the Fact Pack. No remote sources are permitted.
- Identify missing, truncated, duplicate, contradictory, stale, or ambiguous inputs.

- Create the required **offline pre-review checkpoint** under original §4.3 before architecture selection or substantive design decisions. Base it only on the supplied materials. This explicitly substitutes for the original pre-external-research checkpoint because new external research is prohibited in Direct mode; do not pretend that any external source was inspected.

Pass B — Requirements reconstruction

- Reconstruct the complete binding requirement set from the original prompt and this addendum.
- Separate OWNER_HARD, OWNER_PREFERENCE, OWNER_ATTESTED, RATIFIED_FINDING, SOURCE_VERIFIED, DERIVED, DESIGN_JUDGMENT, PROVISIONAL, UNRESOLVED, and OWNER_DECISION_PENDING.
- Identify all updated provisions and the exact original clauses they supersede.
- Preserve every original C/A/D/E/P/G/T-C/V identifier used; check every identifier against its original definition.
- Build one canonical, complete Requirements Register. Use exactly these **11 explicit fields** for every requirement:
  1. `Requirement ID`;
  2. `Requirement / obligation` (full content, not a label only);
  3. `Priority` (`HARD`, `SHOULD`, or `MAY`);
  4. `Authority / origin` (owner decision, original clause, ratified finding, evidence label, or derivation);
  5. `Rationale / source / provenance`;
  6. `Verification method`;
  7. `Acceptance condition` (explicit criterion or `UNRESOLVED` with resolution method);
  8. `Status`;
  9. `Evidence pointer / observed result`;
  10. `Failure response / safe fallback`;
  11. `Regression checks / dependencies / affected requirements`.
- This 11-field schema explicitly resolves the original prompt's internal mismatch between its reference to an 11-field register and its shorter enumerated field list. Preserve all mandatory information from original §6; do not drop a field to force a count.
- Use one table/register, not fragmented primary and auxiliary registers. Additional columns may be added only if justified; the 11 required fields must remain visible.

Pass C — Clean-sheet architecture reconstruction

Before relying on your earlier design, derive a requirements-driven architecture candidate from first principles. At minimum, answer:

- What decisions must the system make, at what cadence, and using which admissible data?
- What strategy/alpha hypothesis is being proposed, why is it appropriate to investigate for Toobit perpetual futures and intraday intent, and what simple baseline would test incremental value?
- What is the minimum state the system must own, and which state is authoritative?
- What needs synchronous correctness, what may be asynchronous, and which transitions must be durable before an external side effect?
- What are the true execution, rate-limit, data, storage, compute, device, and operational envelopes?
- What are the natural component boundaries, and what evidence justifies each one?
- What is the simplest credible architecture that satisfies the hard constraints, and what materially more complex candidates add enough value to justify their costs?

Do not assume a strategy family, optimizer, ML model, feature set, database, message broker, language, framework, microservice topology, cloud provider, FPGA, colocation, FIX, SBE, multicast, or numerical latency target. Do not exclude any such option merely because it is not named here. Derive each inclusion or exclusion positively from requirements and admissible evidence, following original D13 and §9.

Statistical rigor must be used as a verification discipline per D19: propose at least one defensible candidate strategy architecture before concluding that the statistical budget is restrictive. Do not claim that the candidate is profitable, statistically validated, or achievable until the required calculations or tests have actually been performed. Distinguish the strategy/alpha layer from execution algorithms such as TWAP or VWAP.

Pass D — Forensic audit of the previous blueprint

Inspect every material section and artifact of your previous blueprint. For each claim and design decision, determine:

- whether it maps to a real requirement;
- whether the requirement is hard, preferred, delegated, or unresolved;
- whether the supporting fact is genuinely evidenced and correctly scoped;
- whether the chosen solution is positively derived;
- whether alternatives and a simpler baseline were considered where required;
- whether its mathematical logic, units, capacity assumptions, data assumptions, interfaces, and state transitions are coherent;
- whether failure, retry, timeout, recovery, security, and operator-unavailable behavior are specified;
- whether it conflicts with another section, diagram, table, formula, identifier, or owner decision;
- whether it is materially useful or unnecessary complexity;
- whether it can be repaired, must be redesigned, should be removed, or must remain unresolved.

Do not accept an assertion merely because it appears in a polished table or diagram. Do not accept a numerical target merely because it is precise. Do not accept a technology choice merely because it is widely used in institutional trading systems. Precision, sophistication, and length are not evidence.

Create a revision/finding catalog **inside the final design package**, not as a separate standalone audit report. Give every material finding a unique identifier and a defined disposition: `FIXED`, `REDESIGNED`, `REMOVED`, `RETAINED_WITH_JUSTIFICATION`, `UNRESOLVED`, `BLOCKED`, or `NOT_TESTED`. Each finding must identify the previous location, the requirement/evidence basis, severity, rationale, final destination in the revised package, evidence pointer, consequence, and any regression check. Define severity classes before using them:

- `CRITICAL`: violates a hard owner constraint, an essential safety invariant, or could permit uncontrolled risk/incorrect exchange state;
- `MAJOR`: breaks a material capability, correctness property, essential interface, or implementation handoff;
- `MODERATE`: creates meaningful degradation, missing control, ambiguity, or avoidable operational burden but has a bounded safe fallback;
- `MINOR`: a localized noncritical defect in clarity, consistency, or maintainability.

Do not inflate severity to dramatize the review. Do not hide a Critical or Major finding inside a generic paragraph. A finding marked fixed must point to the actual corrected content and a check that verifies the correction.

Pass E — Full-domain audit, including omissions

The audit must be open-ended and must not be confined to this list or to the original prompt's examples. Independently identify any additional discipline, artifact, failure mode, or design dimension required by the actual system. The following is an audit surface to inspect, not a mandatory feature checklist. For each area, include it when material; otherwise record a specific positive reason for exclusion or irrelevance under original D12/D18/D20.

1. **Objectives and system boundary:** measurable objectives, non-goals, capability boundaries, design-only boundaries, responsibilities, trust boundaries, and authority boundaries.
2. **Trading strategy and alpha research:** hypotheses, entry/exit logic, regime handling, multi-timeframe dependencies, market context, no-trade conditions, signal validity, baseline, ablation, negative controls, and strategy lifecycle.
3. **Market and derivatives mechanics:** USDT-M perpetual contract semantics, quantity/notional/contract multiplier, tick/step/minimum values, position mode, margin mode, leverage, maintenance margin, liquidation, mark/index/contract prices, funding, settlement, and contract-specific differences — only to the extent supported by evidence.
4. **Market data:** source coverage, historical availability, WebSocket/REST differences, candle construction, event/availability timestamps, out-of-order messages, gaps, duplicates, stale data, corrections, schema evolution, data lineage, and reproducibility.
5. **Universe and selection:** symbol eligibility, liquidity, volume/depth, spread, volatility, tradability, survivorship/listing changes, correlation/SMT requirements, concentration, refresh cadence, and fail-closed behavior.
6. **Quantitative/statistical methodology:** sample size, effective sample size, serial/cross-sectional dependence, multiple testing, trial budget, selection bias, walk-forward/nested validation, purging/embargo, leakage, robustness, regime shift, uncertainty, calibration, and limits of inference.
7. **Optimization and adaptation:** whether any optimizer is actually needed; separation of signal calibration, portfolio construction, risk authority, and execution tuning; parameter registry; allowed domains; search budget; selection bias; promotion; rollback; drift; and prohibition on changing owner-hard risk limits.
8. **Risk and portfolio:** per-trade and aggregate risk, drawdown, daily loss, effective leverage, exposure/concentration, correlation, margin/liquidation, stress testing, cost uncertainty, missing state, kill switch, veto, latches, resets, and emergency risk reduction.
9. **Execution and OMS:** order intent, validation, order construction, Toobit API mapping, order types/time-in-force, amendment/cancellation, partial fills, duplicates, idempotency, retries, ambiguous outcomes, rate limits, exchange rejection, reconciliation, position accounting, PnL, and external-side-effect durability.
10. **Telegram operator interface:** command catalog, roles, authorization, confirmation, replay protection, message authenticity, rate limiting, idempotency, command preconditions, acknowledgements, refusal messages, audit records, outage behavior, and safe operation when Telegram is unavailable.
11. **Architecture and distributed systems:** logical/process boundaries, synchronous/asynchronous contracts, state ownership, event ordering, persistence, transaction boundaries, queues, consistency, backpressure, split-brain, clock behavior, crash recovery, and deployment topology.
12. **Resource/deployment feasibility:** POCO X6 envelope, Android/Termux constraints, process lifecycle, Doze/background behavior, storage pressure, thermal limits, battery/charging, network interruptions, cloud/VPS/hybrid alternatives, cost, phase placement, and D15 compromise logic.
13. **Security and supply chain:** threat model, secret storage, API-key permissions, least privilege, signing, encryption, dependency pinning, SBOM where justified, update integrity, access control, privilege separation, audit integrity, backup confidentiality, and incident response.
14. **Reliability and operations:** SLO/RTO/RPO as targets rather than measured results, monitoring, alerting, health checks, circuit breakers, degradation modes, restart behavior, backup/restore, disaster recovery, runbooks, and operator unavailability.
15. **Governance and change control:** requirement ownership, evidence provenance, configuration integrity,

versioning, approval, audit trail, state migrations, rollback, policy changes, and traceability.
16. **Verification and delivery:** requirement-to-design traceability, design inspections, arithmetic checks, API contract checks, test specifications, acceptance criteria, implementation handoff, known limitations, and artifact integrity.

A domain or technique not listed above must still be considered if the actual system requires it. Conversely, do not force a named technique or subsystem into the architecture merely because it appears here.

Pass F — Correction and reconstruction

For every defect that can be responsibly resolved from requirements and evidence, correct it in the actual revised design. A list of recommended corrections is not a correction. A statement that a section “will be updated” is not an update. Do not leave the previous blueprint's contradictions in place while describing the proposed fix elsewhere.

Where a correction affects multiple artifacts, update all of them: requirement register, architecture, data model, interfaces, formulas, state machines, diagrams, risk policy, test/verification matrix, operations, runbooks, dependency list, and completeness declaration as applicable.

Where a defect cannot be resolved from the permitted inputs, do not fabricate a resolution. Use the correct status, provide the resolution method and authority, specify safe behavior until resolved, and state the consequence. If the affected feature could create unsafe risk or false state, design it disabled or fail-closed until its prerequisite is established.

Pass G — Adversarial red-team review

Attempt to falsify the revised design. At minimum, challenge:

- the strongest alternative architecture and the simpler baseline;
- every hard owner constraint and every new risk limit;
- every order path, timeout, retry, partial fill, cancel/amend race, and ambiguous exchange response;
- every authoritative-state claim and every reconciliation assumption;
- every data source, timestamp, look-ahead boundary, lineage tag, and cross-venue isolation control;
- every risk veto, kill switch, interlock, latch, reset, and operator authorization;
- every Telegram outage, network outage, process death, storage failure, clock fault, exchange outage, stale feed, and corrupted-state case;
- all resource and capacity estimates, including worst-case bursts and recovery;
- every security trust boundary and privileged action;
- every performance, latency, reliability, and readiness claim;
- every completeness count, identifier, table reference, equation, and cross-reference.

Use fault trees, threat trees, FMEA, invariant checks, boundary-value analysis, concurrency/race analysis, or other methods when they provide real information. State which attacks/failures were actually analyzed and which were not. Do not imply that a paper analysis is a performed operational drill.

Pass H — Regression and coherence verification

After all corrections, inspect the full revised package as one system. Check at least:

- no orphan components, unused data, unjustified dependencies, or unexplained external services;
- no module without a clear state owner and failure policy;
- no identifier used before definition or defined in multiple catalogs;
- no two sections assigning different values or semantics to the same parameter;
- no mismatch between API contract, order state machine, risk engine, reconciliation, and runbooks;
- no mismatch between deployment topology, resource envelope, phase model, and recovery design;
- no research-only cross-venue feed reachable from Paper or Live;
- no path by which a forecast, optimizer, Telegram command, stale cache, or degraded subsystem can bypass a hard risk veto;
- no unsupported Toobit behavior presented as confirmed;
- no unverified value silently promoted to a numeric acceptance threshold;
- no arithmetic mismatch between registers, verification matrix, completeness declaration, or Artifact Registry.

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6. Specific Toobit design controls derived from the Fact Pack

The following reminders do not replace the Fact Pack and must not be treated as additional evidence. They identify design consequences that must be considered and referenced to the exact Fact Pack items.

1. Use the documented REST/WS base URLs, API key header, HMAC-SHA256 signing, lowercase hexadecimal signature, millisecond timestamp, and stated `recvWindow` limits only within the exact scope recorded in the Fact Pack. Do not invent missing canonicalization/signing details.
2. Model the documented REST request-weight and order-request limits separately. The conservative fixed-weight limiter must work without response headers. Do not depend on the headers being present.
3. For new futures orders, preserve the v2 endpoint, required fields, `BUY`/`SELL`, `LONG`/`SHORT`,

quantity/valueQuantity alternatives, permitted time-in-force names, and conditional fields exactly as the Fact Pack lists them. Do not assert support beyond the listed fields or infer undocumented semantics.
4. Treat amend as LIMIT-only according to the Fact Pack. Keep cancel identity rules exact. Do not invent exchange status values beyond `NEW` and `CANCELED` as observed. Local internal states such as `SUBMITTING`, `UNKNOWN`, or `RECONCILING` must be explicitly labeled as local states, not Toobit response enums.
5. A 5XX response, timeout, or ambiguous failure does not prove that an order was rejected. No blind resubmission. Model a durable `UNKNOWN` outcome, lookup/reconciliation, client-order identity, pending exposure, and safe retry rules. The incomplete negative-error catalog remains incomplete.
6. Treat `newClientOrderId` / `origClientOrderId` naming and the `[A-Za-z0-9_-]` constraint as binding. Assume the weakest documented uniqueness scope (per-account) and enforce local uniqueness. Do not reuse an identifier after an ambiguous outcome.
7. Keep futures and spot symbol formats distinct. Use a canonical internal instrument identity with an explicit mapping to the venue symbol; do not silently assume `BTCUSDT` equals `BTC-SWAP-USDT`.
8. Treat `exchangeInfo` as the source for startup discovery of per-symbol filters. The actual minQty, stepSize, tickSize, and per-symbol leverage values are not included in the Fact Pack. No trade may be designed around invented filters.
9. Preserve the 14-timeframe ratified capability ceiling while separately recording the narrower WebSocket list and unresolved REST/WS availability. Use the conservative 1,000-kline return bound where the documented upper limit conflicts. Do not treat 3,500 as guaranteed history.
10. Use the stated WebSocket heartbeat semantics. Specify disconnect detection, stale-feed behavior, resubscription, gap repair, duplicate handling, and whether a reconstructed candle is provisional or final.
11. Funding settlement frequency varies by contract according to the Fact Pack. The provisional funding caps are not guaranteed limits. Do not assume funding cost is bounded by the observed C7 clamp.
12. Use 0.06% as the conservative taker-fee default and report the unresolved 0.05% alternative as sensitivity. Do not omit maker/taker assumptions or pretend the owner's actual fee tier has been verified.
13. The BTC leverage example and provisional altcoin values do not define a universal exchange maximum. The owner’s effective-leverage ceiling is separately 5×.
14. Do not assume Cross/Isolated margin behavior is confirmed. Preserve its `PROVISIONAL` status and define the safe dependency on actual supported account/symbol configuration.
15. Do not treat the Fact Pack's partial symbol list as the current universe. Derive the universe under D14 and require runtime exchange-symbol/filter discovery in the eventual implementation design.
16. Do not assume historical OI, reproducible order-book depth, or conflicted mark-price semantics. A feature that depends on an unresolved data class must have a safe fallback or remain disabled.
17. Paper mode is an internal simulation design, not proof of a Toobit testnet. The Fact Pack's statement about no sandbox/testnet wired in CCXT must not be overgeneralized beyond its wording.

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7. Architecture and quantitative correctness rules

7.1 Requirements-first, not template-first

Follow original D11, D13, D18, and D12. Architecture dimensions, module count, deployment location, data stores, messaging, framework, language, models, optimizers, and document count must be derived. Do not use institutional-trading terminology, service count, low-latency jargon, or fashionable infrastructure as proxies for quality.

7.2 Strategy, execution, and optimization are distinct

A trading strategy/alpha model defines when and why to enter, hold, reduce, or exit a position. Execution algorithms determine how a valid order intent is implemented. TWAP/VWAP/POV/implementation shortfall are not substitutes for the strategy itself.

Provide a specific candidate strategy architecture or clearly defined research hypothesis that meets the original requirements, plus a defensible simple baseline. State all required inputs, signal semantics, timeframes, entry/exit/no-trade logic, holding policy, and failure conditions. Keep the strategy `NOT TESTED` where it has not been backtested. Do not claim an expected return or statistical pass without executed supporting evidence.

An optimizer is not mandatory merely because the owner or prior blueprint mentioned one. Derive whether optimization adds material value, keep trial budgets explicit, and distinguish forecast calibration, strategy selection, portfolio construction, risk control, and execution parameter tuning. The hard D-RISK limits are never tunable.

7.3 Risk authority is independent

Define one authoritative risk-policy catalog. Specify pre-trade and continuous risk checks, exact order of evaluation, data freshness, authoritative equity/position inputs, veto behavior, risk-reducing actions, Kill Switch semantics, latches, acknowledgement, re-arm, and audit events.

The six Live interlocks are mandatory and non-bypassable for entry into Live mode and for every risk-increasing action to the extent required by its safety purpose. Specify the action taxonomy so a failure does not accidentally block a necessary risk-reducing action or permit an unsafe one. Do not invent exchange support for reduce-only, stop, or emergency semantics.

A missing/corrupt risk-state record, unknown position, stale equity, unverified open orders, clock fault,

data-integrity fault, config-integrity fault, or unresolved exchange outcome must not silently pass a risk-increasing gate.

7.4 Durable order side effects and ambiguous outcomes

Specify the exact moment at which order intent, client identity, risk reservation, and submission attempt become durable relative to sending an external request. Distinguish at least:

- local intent not yet sent;
- send attempted;
- acknowledgement observed;
- rejection observed;
- partial execution;
- full execution;
- cancellation requested;
- cancellation confirmed;
- amendment requested/confirmed;
- timeout or 5XX with unknown outcome;
- reconciliation in progress;
- state mismatch or terminal unresolved state.

Only use Toobit-confirmed exchange states where supplied. Internal states must be visibly identified as local abstractions. Define idempotency, deduplication, retry eligibility, retry limits, and position/order reconciliation without assuming exactly-once delivery from an external API.

7.5 Paper/Live mode semantics

The owner may select Paper or Live through Telegram without waiting periods or profitability thresholds. The six interlocks remain mandatory. Define configuration separation, state isolation, order-sending boundary, mode-switch transaction, pending-order behavior, open-position handling, and audit event. Make it impossible for Paper to submit real orders and impossible for Live to consume cross-venue research-only data.

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8. Deliverable-content policy: comprehensive without mechanical bloat

The final package must answer every material design question raised by the original requirements and by this audit. It must not be restricted to the headings in the original blueprint, this prompt, or any standard architecture template. Add any missing artifact, diagram, catalog, contract, formula, control, or specification that is materially necessary.

However, do **not** mechanically force every example below into the design. Original D12/D18 control: a checklist is an omission-audit tool, not an architecture template. Include a deliverable when it answers a real requirement or enables verification/implementation. If a requested or conventional artifact is not relevant, exclude it with a positive reason, evidence/status, and consequence. For example, do not force VaR/CVaR, Smart Order Routing, microservices, Kubernetes, a dashboard, FIX/SBE, FPGA, colocation, or a particular CI/CD release method unless the requirements and evidence justify it. If SOR is relevant, constrain it to actual Toobit-supported routing/order behavior; do not imply multi-venue routing for Live.

Where material, the package may need to cover:

- system context and architecture diagrams;
- component responsibilities and state ownership;
- sequence and data-flow diagrams;
- requirements traceability and architecture decision records;
- exchange/API and internal interface contracts;
- instrument, order, position, equity, risk, and event schemas;
- data ingestion, quality, lineage, retention, and storage design;
- strategy/signal and execution specifications;
- risk, portfolio, position, and order lifecycle state machines;
- Telegram commands, permissions, confirmations, and refusal behavior;
- security/threat model and secret lifecycle;
- resource and deployment topology comparison;
- SLO/RTO/RPO design targets with explicit non-measured status;
- monitoring, alerting, runbooks, recovery, and disaster-recovery design;
- test strategy, acceptance criteria, and Definition of Done;
- future implementation dependency order and handoff boundaries;
- dependency inventory, risk register, change controls, and artifact registry.

Every included artifact must be complete for its declared purpose. No decorative artifact counts as coverage of an underlying behavior.

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9. Design freeze definition and status rules

For this engagement, **Frozen** means every material design item has a defined content, rationale, authoritative source or evidence label, status, evidence pointer, consequence, and safe handling. A frozen design may contain `UNRESOLVED`, `BLOCKED`, `NOT TESTED`, or `PROVISIONAL` items where the uncertainty is explicitly bounded and the dependent capability is safely disabled, constrained, or fail-closed.

Frozen does **not** mean implemented, operationally proven, profitable, validated on historical data, Paper-tested, production-ready, or authorized for Live trading.

Use acceptance statuses consistently:

- `PASS`: observed evidence satisfies the declared criterion;
- `FAIL`: observed evidence violates the criterion;
- `BLOCKED`: a required source, dependency, resource, or authorization is unavailable;
- `NOT TESTED`: the test or measurement has not been executed;
- `UNRESOLVED`: the underlying fact or decision is not established or is conflicting.

Use evidence labels from original §2 for source/authority classification. `PROVISIONAL` and `DESIGN_JUDGMENT` are not substitutes for test statuses. A design proposal may be `PROVISIONAL`; an acceptance test that has not been run is `NOT TESTED`.

Do not declare the entire package `NO-GO` merely because a noncritical or safely contained item is `UNRESOLVED`. Record the affected requirement, disable or fail-close the dependent capability, state the resolution path and consequence, and continue. A `NO-GO` conclusion is permitted only where a genuine, material contradiction between binding requirements or an unavoidable essential safety conflict makes a compliant design impossible; cite the exact clauses and prove why no safe alternative or containment exists.

Conversely, this rule does not permit unsafe acceptance. If an unresolved item affects an essential safety invariant and cannot be safely contained, the dependent Live/risk-increasing capability must remain disabled or blocked. Never convert this into a claim that the system is ready.

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10. Completeness, identifiers, verification, and delivery

Apply original §2.2 (D20/D21/D22), §6, §11, §12, and §13 in full. The following clarifications are mandatory for this refinement:

1. Every material item must contain actual content, rationale, status, evidence pointer, and consequence. A label, heading, reference, or promise to revise later is not content.
2. Do not use summaries, abbreviations, unexplained acronyms, omitted ranges, ellipses, “etc.”, “see above”, “TBD”, “to be covered later”, “for brevity”, or equivalent compression in place of required content. Define every acronym at first use. No false claim of completeness.
3. Keep one canonical catalog for every repeatedly used identifier, requirement ID, status class, order state, error state, event type, API field mapping, risk limit, configuration key, alert, and artifact. No dangling or duplicate definitions.
4. The Requirements Register must be a single table using the 11 fields specified in Pass B.
5. The Verification Matrix must contain exactly one row for every requirement-register entry, no missing or phantom requirement rows. Multiple verification procedures may be enumerated inside the corresponding requirement row. State the counts and verify equality arithmetically.
6. The completeness declaration must use the original §2.2.5 structure, count every item exactly once, and show a reproducible arithmetic reconciliation. Distinguish actual `PASS`, `FAIL`, `BLOCKED`, `NOT TESTED`, and `UNRESOLVED` statuses from items intentionally excluded or not relevant. Do not double-count any item.
7. Reproduce at least three substantive arithmetic walks from the design, with inputs, units, formula, intermediate steps, and final result. Include at least one risk calculation, one capacity/resource or rate-limit calculation, and one statistical/trial-budget or completeness calculation when such calculations are supported by available inputs. If an item cannot be calculated, state why and label it accordingly; do not invent an input.
8. Enumerate original V-1 through V-12 individually in a table. Each row must include the check, result, evidence pointer, and corrective action where applicable. A blanket claim that all checks passed is prohibited.
9. Maintain an Artifact Registry per original §2.2.8, updated incrementally. Include artifact ID, artifact name, phase produced, message/part reference, status (`DELIVERED`, `DEFERRED`, or `BLOCKED`), and concise content description. A named artifact must actually be present in the delivered output or linked file.
10. Include an adversarial-review change-propagation table inside the final design package. For each material finding that led to a change, identify finding ID, changed section/artifact, exact location, description of the new content, and the verification that the correction is present. Do not substitute a separate audit report for the actual corrected design.
11. Preserve the seven independent dimensions in original §11.3: design completeness, source/evidence review, research/backtest evidence, implementation correctness, Paper behavior, operational readiness, and Live authorization. Report each separately.
12. SLO, RTO, RPO, latency, throughput, capacity, and resource values are design targets or estimates unless actually measured in this engagement. Label them `DESIGN_JUDGMENT`, `PROVISIONAL`, or `NOT TESTED` as appropriate. Never represent them as benchmark results.
13. Do not claim actual implementation, exchange connectivity, historical backtest, Paper operation, production deployment, operational rehearsal, or Live readiness. The test plan can be fully specified while the test execution remains `NOT TESTED` / `OUT OF SCOPE`.

14. If output exceeds one message, obey D22: announce the part plan and honest length-check method, use `Part 1/N` through `Part N/N`, complete every part for its declared scope, cross-reference accurately, and include the required completeness declaration in every part and the final overall declaration. Do not claim that future parts have already been delivered. If the environment does not permit multi-message delivery, disclose that as `BLOCKED` and deliver the largest complete unit possible without pretending the package is complete.
15. Follow incremental persistence in original §2.2.8. Do not hold material findings only in working memory. Deliver phase artifacts and update the registry as required.
16. Owner-facing progress/status notes and the short final note must be Persian; the principal design package must be professional English Markdown.

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11. Required final package behavior

The primary final output is the **revised, fully integrated design package**, not a standalone critique of your prior answer. Do not replace the requested deliverable with a list of recommendations, a prompt, a review essay, or an audit summary.

All material audit findings must be reflected in the revised design itself. Include only the integrated revision/finding catalog and change-propagation table needed to make the corrections auditable. The package must be coherent as a single specification: no section may rely on an unresolved contradiction that is hidden elsewhere; no requirement may lack a corresponding design disposition; no interface may be left for an engineering team to invent; and no state transition may be described only in prose when an explicit contract or state model is necessary to remove ambiguity.

Before final delivery, verify the entire package against:

- all original requirements, constraints, evidence labels, precedence rules, and D11–D22 obligations;
- the six owner-hard D-RISK decisions and their cross-interactions;
- the six non-bypassable Live interlocks;
- the Telegram-only interface requirement;
- the Toobit-only Live execution boundary and cross-venue research/backtest-only isolation;
- the D14 universe derivation and C19 non-universe rule;
- D15 topology enumeration and envelope-first compromise order;
- D16 phase derivation and no-wait/no-profit-hurdle Paper/Live selector;
- D19 candidate strategy-before-statistical-rejection rule;
- C20–C24 statistical and trial-budget controls;
- every Toobit Fact Pack fact, status, conflict, unresolved item, and conservative branch;
- all data, risk, execution, OMS, reconciliation, reliability, security, Telegram, mobile, deployment, testing, and operations dependencies actually material to the design;
- every identifier, cross-reference, formula, register count, matrix count, completeness calculation, and artifact declaration.

Correct every defect that can be resolved using the permitted information. For anything that cannot be resolved, disclose the exact missing evidence/authority, safe interim behavior, dependency, and consequence. Do not issue an unqualified claim that the blueprint is “perfect”, “complete in every possible sense”, “production ready”, “profitable”, or “fully validated”.

Final instruction: Execute the reassessment and reconstruction now. Deliver the strongest defensible revised architecture that the original master prompt, the binding owner addendum, the Toobit Fact Pack, and your actual available capabilities permit. Do not stop at criticism. Do not merely suggest fixes. Make the fixes in the design. Preserve all unresolved facts honestly and safely. The final design package — not an assertion of excellence — must demonstrate the result.

Direct-mode continuation rule: Do not stop, defer the task, or return a generic statement that internet access is required. Do not ask for browsing or Agent mode. Complete every portion that can be supported by the closed-world inputs and offline reasoning. Isolate externally dependent items, mark their source-verification step `BLOCKED` and/or their unexecuted test `NOT TESTED`, retain the underlying fact as `UNRESOLVED` where appropriate, give a concrete future evidence/measurement path and safe interim behavior, and continue. Never represent the absence of internet as evidence that a missing fact is true or false.

TOOBIT FACT PACK — Extracted Data for Design Engagement
Source: api-docs.toobit.com, CCXT docs, public exchange announcements
Extraction date: 2026-11-02 (UTC)
Status: SOURCE_VERIFIED where cited; otherwise PROVISIONAL or UNRESOLVED as marked.
Warning: This is a snapshot, not a live contract. Items marked UNRESOLVED must
be treated as UNRESOLVED in the design, not silently filled.

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1. API Base and Connectivity

| Property | Value | Status |

|---|---|---|
REST base URL	`https://api.toobit.com`	SOURCE_VERIFIED
WebSocket base URL	`wss://stream.toobit.com`	SOURCE_VERIFIED
API key header	`X-BB-APIKEY`	SOURCE_VERIFIED
Signing method	HMAC SHA256	SOURCE_VERIFIED
Signature case	lowercase hex	SOURCE_VERIFIED
Timestamp unit	milliseconds	SOURCE_VERIFIED
recvWindow default	5000 ms	SOURCE_VERIFIED
recvWindow max	60000 ms	SOURCE_VERIFIED
Max future clock skew	1000 ms (server rejects timestamp > serverTime + 1000)	SOURCE_VERIFIED
Sandbox / testnet	None wired in CCXT	SOURCE_VERIFIED
Official SDK	None published; signing examples only	SOURCE_VERIFIED

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2. Rate Limits (current documented state)

Type	Limit	HTTP on violation	Error code
REQUEST_WEIGHT	3000 weight per 1 minute	429	-1003 TOO_MANY_REQUESTS
ORDERS	60 order requests per 2 seconds	429	-1015 TOO_MANY_ORDERS

Rate-limit response headers are documented as present (X-Api-Limit-Status, X-Api-Limit,
X-Api-Limit-Reset-Timestamp). However, the master prompt sent in 2 parts ratified finding C4 says these
headers were absent in the owner's tested environment. Resolution: treat as UNRESOLVED with two
branches — (a) if headers present, use delta; (b) if absent, use conservative fixed-weight
accounting. The design MUST implement (b) as the safe baseline and MAY use (a) if
headers are observed.

Endpoint weights (examples):
- `/api/v1/time`, `/api/v1/ping`, `/api/v1/exchangeInfo`: 1
- `/api/v2/futures/order` (POST): 1
- `/api/v2/futures/order` (DELETE): 1
- `/api/v2/futures/batch-orders` (POST): 2
- `/api/v2/futures/batch-orders` (DELETE): 3
- `/api/v2/futures/order/update` (POST): 2
- `quote/v1/ticker/24hr`: 40
- `api/v1/futures/positions`: 8.35

---

3. Futures Order Lifecycle (v2, USDT-M)

New regular order
- Endpoint: `POST /api/v2/futures/order`
- Weight: 1
- Mandatory fields: `symbol`, `side` (BUY/SELL), `positionSide` (LONG/SHORT),
  `type` (LIMIT/MARKET), `newClientOrderId`, `timestamp`
- Quantity: `quantity` (contracts) or `valueQuantity` (USDT notional), at least one non-zero
- TimeInForce: GTC (default), FOK, IOC, POST_ONLY
- Conditional fields: `price` (required for LIMIT), `stopPrice`, `takeProfit`, `stopLoss`,
  `tpTriggerBy`/`slTriggerBy` (CONTRACT_PRICE or MARK_PRICE), `tpOrderType`/`slOrderType`
  (MARKET or LIMIT), `stopOrderType` (TAKE_PROFIT/STOP_LOSS), `stopType` (FIXED_STOP/TRAILING_STOP)

Update order (amend)
- Endpoint: `POST /api/v2/futures/order/update`
- Weight: 2
- LIMIT only; amend `price` / `quantity` by `orderId` or `origClientOrderId`

Cancel order
- Endpoint: `DELETE /api/v2/futures/order`
- Weight: 1
- Identify by `orderId` or `origClientOrderId` (mutually exclusive)

Batch operations
- `POST /api/v2/futures/batch-orders`: Weight 2, JSON array body
- `DELETE /api/v2/futures/batch-orders`: Weight 3

Order response fields (observed shape)
`time`, `updateTime`, `orderId`, `clientOrderId`, `symbol`, `price`, `leverage`,
`origQty`, `executedQty`, `avgPrice`, `marginLocked`, `type`, `side`, `positionSide`,
`timeInForce`, `status`, `contractMultiplier`

Status values observed: NEW, CANCELED. Other statuses not confirmed in extracted docs —
treat as UNRESOLVED if the design requires the full state set.

Symbol format
Futures symbols use format like `BTC-SWAP-USDT`, `ETH-SWAP-USDT` in v2 responses.
Spot symbols use `BTCUSDT`. The design MUST NOT assume a single symbol format
across spot and futures.

clientOrderId
- Parameter name: `newClientOrderId` for creation; `origClientOrderId` for lookup/cancel
- Character set: master prompt sent in 2 parts C11 says `[A-Za-z0-9_-]` — this is the binding constraint
- The docs show examples like `test1115`, `c1`, numeric strings — all within the set
- Uniqueness scope: UNRESOLVED (whether unique per-symbol, per-account, or globally
  is not stated in extracted docs). Design must assume weakest (per-account) and enforce
  its own uniqueness.

---

4. Error Codes (partial — full list NOT extracted)

Code	Meaning	Source
-1003	TOO_MANY_REQUESTS	SOURCE_VERIFIED
-1015	TOO_MANY_ORDERS	SOURCE_VERIFIED
-2011	Unknown order (seen in batch response example)	PROVISIONAL
-2015	Authentication error (mentioned in skill docs)	PROVISIONAL
429	HTTP rate-limit	SOURCE_VERIFIED
403	WAF limit	SOURCE_VERIFIED
5XX	Internal error, execution status UNKNOWN	SOURCE_VERIFIED

UNRESOLVED: Full negative-code catalog is not in the extracted data. The design
MUST NOT invent error codes. If a state transition depends on a specific code,
mark it UNRESOLVED with a resolution path (query the full error-code page at
api-docs.toobit.com/api/usdt-m-error-codes.html).

C5 note: `-1007 TIMEOUT` is documented in master prompt sent in 2 parts as ambiguous. It does not
appear in the extracted partial list. Treat it as UNRESOLVED but honor the master prompt sent in 2 parts
rule: timeout never proves rejection, no automatic resubmit until reconciliation.

---

5. Klines and Market Data

Property	Value	Status
REST kline endpoint	`GET /quote/v1/klines`	SOURCE_VERIFIED
Mark-price kline endpoint	`GET /quote/v1/markPrice/klines`	SOURCE_VERIFIED
Index kline endpoint	`GET /quote/v1/index/klines`	SOURCE_VERIFIED
WebSocket kline topic	`kline_$interval`	SOURCE_VERIFIED
WebSocket mark-price kline topic	`markPriceKline_$interval`	SOURCE_VERIFIED
Kline return upper limit	2000 (per one endpoint docs) / 1000 (per another)	PROVISIONAL — two figures observed; design must use the conservative 1000 unless verified
Intervals (WebSocket)	1m, 5m, 15m, 30m, 1h, 2h, 6h, 12h, 1d, 1w, 1M	SOURCE_VERIFIED for this list
master prompt sent in 2 parts C1 list	14 timeframes including 3m and 8h	RATIFIED_FINDING from master prompt sent in 2 parts — the WebSocket extracted list is narrower; REST may support more
Depth endpoint	`GET /quote/v1/depth`	SOURCE_VERIFIED
24hr ticker	`GET /quote/v1/ticker/24hr` (weight 40)	SOURCE_VERIFIED

C1/C2 conflict: master prompt sent in 2 parts says 14 timeframes are supported (C1) and 3,500 candles
is a cap (C2). The extracted WebSocket docs show 11 intervals (missing 3m, 4h, 8h).
This is a DOC_CONFLICT: master prompt sent in 2 parts is the ratified finding; the WebSocket doc may be
stale or may cover only a subset. The design MUST treat the 14-timeframe list as
RATIFIED (per master prompt sent in 2 parts C1) and treat 3,500 as a cap not a guarantee (C2). Do not
downgrade the universe to 11 timeframes based on this conflict — instead, record
the conflict and proceed with the C1 list as the ceiling.

---

6. Funding

Property	Value	Status
Funding settlement frequency	every 8 hours (standard), some contracts every 4 hours	SOURCE_VERIFIED
Funding rate cap (observed range)	±2.0000% to ±3.0000% depending on contract	PROVISIONAL — varies per contract
Funding rate endpoint	`GET /api/v1/futures/fundingRate`	SOURCE_VERIFIED
History funding rate endpoint	`GET /api/v1/futures/historyFundingRate`	SOURCE_VERIFIED

C7 note: master prompt sent in 2 parts says a funding clamp of −30.0000 bp was observed and the alleged

±24 bp cap was refuted. The extracted data shows ±2%–3% caps, which is consistent
with the refutation. Treat funding as tail risk, not bounded cost.

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7. Fees

Product	Maker	Taker	Status
Futures (USDT-M perpetual) — base VIP0	0.02% (2 bps)	0.06% (6 bps)	SOURCE_VERIFIED (multiple sources agree)
Futures — alternative observed	—	0.05%	UNRESOLVED (master prompt sent in 2 parts C6: 0.05% documented vs 0.06% observed)
Spot — standard	0.075%	0.10%	SOURCE_VERIFIED
Spot — promotional	0%	0%	PROVISIONAL — waived on selected pairs since June 2026

C6 resolution for design: use 0.06% taker as the conservative default. Do not
present 0.05% as settled. Show sensitivity across both.

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8. Leverage

Property	Value	Status
BTCUSDT perpetual max leverage	up to 200× (tier 1: notional 0–1 BTC, maintenance margin 0.25%)	SOURCE_VERIFIED from Toobit support announcement
Other perpetuals max leverage	50× typical, 75× for some listed contracts	PROVISIONAL — varies per contract
Risk-limit tiers endpoint	`GET /api/v1/futures/riskLimits`	SOURCE_VERIFIED
master prompt sent in 2 parts C3 ladder	13 tiers down to 1.0×	RATIFIED_FINDING

C3 note: master prompt sent in 2 parts says the 13-tier ladder is the ratified finding, but the extracted
data shows different tier structures (e.g., 200× at tier 1 for BTC, 125× in another
source, 50× for many alts). The design MUST NOT infer a single maximum from any one
tier. Treat the maximum permitted leverage as UNRESOLVED for each symbol; the ladder
is tier information, not a system risk limit.

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9. Contract Specifications

Property	Value	Status
contractMultiplier (observed in response)	"0.001" for BTC-SWAP-USDT, "0.01" for ETH-SWAP-USDT	SOURCE_VERIFIED from response example
Min order / qty step / tick size	Per-symbol, available via `/api/v1/exchangeInfo`	SOURCE_VERIFIED that the endpoint exists; the values for each symbol are NOT extracted
Margin modes	Cross / Isolated (standard on Toobit futures)	PROVISIONAL — not confirmed in extracted docs but widely documented

UNRESOLVED: Full per-symbol contract specification table (minQty, stepSize, tickSize,
maxLeverage) is not extracted. The design MUST treat the universe's per-symbol filters
as UNRESOLVED and specify that the implementation must fetch them at startup from
`/api/v1/exchangeInfo` and validate against them. Do not invent filters.

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10. Symbol Universe (partial)

The following perpetual contract families are mentioned in extracted sources:
BTCUSDT, ETHUSDT, SOLUSDT, BNBUSDT, XRPUSDT, ADAUSDT, DOGEUSDT, AVAXUSDT, LINKUSDT,
LTCUSDT, BALUSDT, XVSUSDT, ONGUSDT, WAXPUSDT, IOSTUSDT, RLCUSDT, JOEUSDT, CTSIUSDT,
TRUUSDT, HIGHUSDT, 1000WHYUSDT, 1000CHEEMSUSDT, MLNUSDT, FUNUSDT, NEARUSDT, SANDUSDT,
MATICUSDT, MANAUSDT, ATOMUSDT, APTUSDT, FTMUSDT.

Status: PROVISIONAL — this is a partial list from announcements, NOT the current
full contract list. The design MUST NOT treat this as the universe. The full current
list must be fetched at runtime from `/api/v1/exchangeInfo` (public, no auth required).

C19 note: The 10 symbols in master prompt sent in 2 parts C19 (BTC, ETH, SOL, BNB, XRP, ADA, DOGE,
AVAX, LINK, LTC) are a measurement set, not a universe. They are incidental if they
survive the D14 filters.

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11. WebSocket Streams (summary)

Topic	Description
`realtimes`	Real-time price updates
`trade`	Per-trade stream
`kline_$interval`	Candlestick updates
`depth`	Order book depth
`markPrice`	Mark price
`markPriceKline_$interval`	Mark-price candlestick
`index`	Index price
`indexKline_$interval`	Index-price candlestick

Heartbeat: client sends `{"ping": <timestamp>}`, server responds with `{"pong": <timestamp>}`.
Server disconnects within 5 minutes if no ping. Design MUST implement heartbeat.

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12. What This Fact Pack Does NOT Establish

The following remain UNRESOLVED and MUST be labeled as such in the design:

- Full error-code catalog
- Per-symbol minQty / stepSize / tickSize table
- Exact K-line historical depth per symbol per timeframe (master prompt sent in 2 parts C2 says 3,500 cap, 1,774 daily sample observed; do not assume 3,500 is guaranteed)
- Order-book depth reproducibility (master prompt sent in 2 parts C8: NON_REPRODUCIBLE)
- markPrice / klines.change field semantics (master prompt sent in 2 parts C13: CONFLICTED)
- Historical Open Interest (master prompt sent in 2 parts C26: UNRESOLVED)
- Whether clientOrderId uniqueness is per-account or per-symbol
- Whether the full 14-timeframe set is available via REST or only a subset via WebSocket
- Whether rate-limit response headers are present in the owner's actual environment (master prompt sent in 2 parts C4 says absent; docs say present; DOC_CONFLICT — use conservative branch)
- Maximum safe outage / exchange retention period (master prompt sent in 2 parts C12: PROVISIONAL 3-day, class definition undefined)

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13. How to Use This Fact Pack

1. Treat every SOURCE_VERIFIED item as evidence with an inspectable source.
2. Treat every PROVISIONAL item as a working value with a stated resolution path.
3. Treat every UNRESOLVED item as exactly that — do not fill it with invented numbers.
4. Where master prompt sent in 2 parts and this Fact Pack conflict, master prompt sent in 2 parts's RATIFIED_FINDING wins
   (per master prompt sent in 2 parts §2 precedence rule 3), unless the conflict is a DOC_CONFLICT that master prompt sent in 2 parts itself flags (C4, C13). For those, use the conservative branch and label.
5. Every design choice that depends on a Fact Pack item must cite the item's status.